
Problem Statement

Annexure A – Scenario Based Assignment
All Five Questions (Q1 – Q5)

Course Name	Artificial Intelligence
Course Code	CSA17
Assessment	Assessment Tool 2 – Annexure A (Scenario Based Assignment)
Course Outcome	CO2 – Search & Game-Solving Techniques (BL3)
Weightage	50%
Duration	1 Hour

Declaration: I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment, in accordance with the Code of Conduct in Annexure A.

Name / Reg. No.: _____

Signature of the Student: _____

Code of Conduct

I, _____ (Name / Reg. No.), certify that the responses submitted against this problem statement represent my original work, and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Question 1: Emergency Drone Delivery in a Flood-Affected Region

Relevant Techniques: Greedy Best-First Search vs. A* Search

Scenario: A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic — roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to a heuristic estimate (straight-line distance), partial real-time updates about blocked paths, and variable movement costs depending on terrain and weather. Due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- Behaviour of Greedy Best-First Search under uncertainty
- How A* balances cost and heuristic
- iii. Impact of heuristic accuracy on both algorithms
- Effect of dynamic changes (blocked paths) on search performance
- Recommended algorithm and justification

Question 2: Smart City Traffic Signal Optimization

Relevant Techniques: Hill Climbing vs. Simulated Annealing / Genetic Algorithms

Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize total waiting time, fuel consumption and traffic congestion. The system must continuously adjust signals based on traffic flow. However, the search space is extremely large, traffic patterns change dynamically, and the system may get stuck in locally optimal solutions.

Tasks:

- Problem formulation and why traditional search is inefficient
- Hill Climbing behaviour and its limitations
- Local maxima, plateaus and ridges — real-world interpretation
- iii. How Simulated Annealing / Genetic Algorithms overcome these limitations
- Recommended approach and justification

Question 3: Autonomous Mars Rover Exploration

Relevant Techniques: Online Search Agent under Partial Observability

Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability, and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- Why an online search agent is required instead of offline search
- Challenges of partial observability and dynamic environments
- iii. How the rover builds and updates its internal model
- Online search compared with uninformed and informed offline search
- Justification of the most suitable approach

Question 4: University Exam Timetable Generation

Relevant Techniques: Constraint Satisfaction Problem — Backtracking with Forward Checking

Scenario: A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: no student should have overlapping exams, a limited number of exam halls exist, certain subjects must be scheduled before others, and faculty availability and special student accommodations must be respected. The complexity increases as the number of courses and students grows.

Tasks:

- CSP formulation
- Variables, domains and constraints — detailed definition
- How backtracking search works in this scenario
- iii. How forward checking improves efficiency
- Real-world challenges and justification of the best strategy for scalability

Question 5: Strategic Game AI for Real-Time Decision Making

Relevant Techniques: Minimax and Alpha-Beta Pruning

Scenario: A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, make decisions within strict time limits, and evaluate complex game states with multiple possible moves. The decision tree is extremely large, making computation expensive.

Tasks:

- How Minimax models this problem
- Computational challenges of large game trees
- iii. How Alpha-Beta pruning improves efficiency
- Designing an evaluation function and justifying the chosen factors
- Recommended strategy for balancing optimality and real-time performance